

FOR SECTION I, BUBBLE ALL ANSWERS ON SCANTRON.
FOR SECTION II, WRITE ALL ANSWERS ON THIS EXAM.

20 pts total

SECTION I: MULTIPLE CHOICE

SELECT SOLELY ONE ANSWER PER QUESTION ON THE PROVIDED SCANTRON.

15 QUESTIONS – 15 POINTS – 15 MINUTES

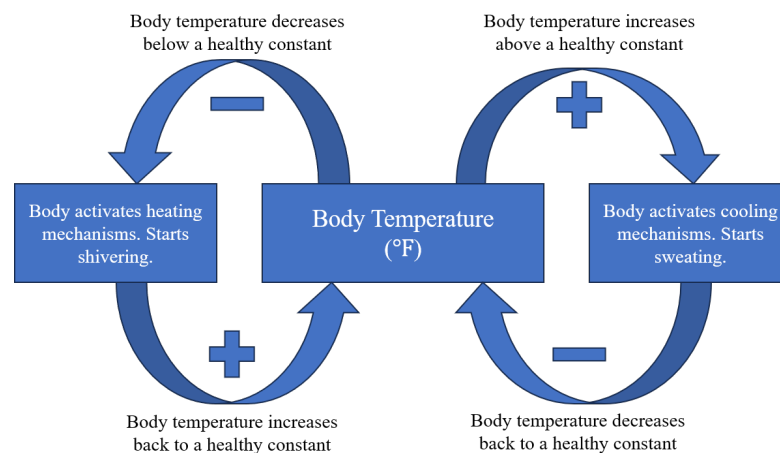
pacing recommendation: 1 minute / question

Ecology 1:

1. Which of the following are TRUE in relation to the diagram below?

- I: The diagram depicts a positive feedback mechanism.
- II: The diagram depicts a negative feedback mechanism.
- III: The diagram depicts homeostasis incorrectly.
- IV: The diagram does not represent homeostasis in any way.

- A. I and III
- B. II and III**
- C. I and IV
- D. II only



2. Which of the following statements is TRUE in regard to homeostasis?

- A. Homeostasis involves maintaining relatively stable internal conditions that fluctuate within a bounded range, often differing from the external environment.**
- B. Homeostasis ensures organisms are in complete equilibrium with their external environment, maintaining identical internal and external conditions.
- C. Homeostasis refers to a state where internal conditions are unstable but in equilibrium with the external environment's fluctuations.
- D. In general, homeostasis is maintained through positive feedback mechanisms.

3. What is the ultimate source of energy for all life on Earth?

The ultimate source of energy is the sun!!

- A. Decomposers
- B. Heterotrophs
- C. Autotrophs
- D. None of the above**

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4. As energy is transferred throughout the food chain, which of the following occurs?
- A. Energy is converted into organic matter.
 - B. Energy is released as heat.
 - C. Energy is destroyed.
 - D. Organic matter is converted into energy.
5. Which of the following are utilized by producers to obtain chemical potential energy?
- I: Photosynthesis
 - II: Cellular Respiration
 - III: Chemosynthesis
- Chemosynthesis is a similar pathway to photosynthesis (used by organisms who don't have access to oxygen)!!
- A. I and III – since producers can obtain chemical potential energy under exposure to sunlight and without in some cases.
 - B. II only – since cellular respiration breaks down glucose and other biomolecules in order to obtain usable energy in the form of ATP.
 - C. I only – since only autotrophs can perform chemosynthesis
 - D. None of the above
6. Which of the following is likely to happen if producers were fully removed from a particular ecosystem?
- I: Heterotroph populations would likely increase.
 - II: Heterotroph populations would likely decline.
 - III: Apex predator populations would likely decline.
 - IV: Secondary consumer populations would likely increase since secondary consumers are not directly dependent upon producers for energy.
- A. I and IV
 - B. II and IV
 - C. II and III
 - D. None of the above
7. Which of the following is likely to happen if secondary consumers were fully removed from a particular ecosystem?
- I: Autotroph populations would immediately seriously drop.
 - II: Primary consumer populations would immediately seriously drop.
 - III: Decomposer populations would immediately seriously drop.
 - IV: Tertiary consumer populations would soon seriously drop.
- A. I and IV
 - B. III and IV
 - C. I, II, III, and IV
 - D. None of the above

8. Which of the following is an abiotic factor?

- A. Humidity
- B. Fallen Tree Leaves
- C. Gopher Snakes
- D. Both A and B

Ecology 2:

9. Fill In The Blank: Expansive slight differences in organisms of the same species in a particular ecosystem indicates high _____.

- A. Genetic Diversity
- B. Species Diversity
- C. Ecosystem Diversity
- D. Both A and B

10. Which of the following are provisioning ecosystem services?

- I: Cycling Matter
 - II: Food
 - III: Shelter
 - IV: Flood Control
- For students not in Ms. Lerner's classes, you do not have to know the subtypes of the ecosystem services, but you do have to know what an ecosystem service is in general!

- A. I, II, and III
- B. IV only
- C. II and III
- D. I only

11. Which of the following is an example of a keystone species?

- A. **Gray Wolf:** Controls herbivore populations, preventing overgrazing and thus protecting plant species diversity in forests.
- B. **White-Footed Mouse:** Forages on seeds and small insects, contributing to nutrient cycling in forest ecosystems.
- C. **Common Starling:** Feeds on fruits and insects, participating in seed dispersal and local food webs.
- D. None of the above

12. Which of the following best describes conservation science?

- A. The study of maximizing biodiversity by increasing the total number of species in an ecosystem regardless of their impact.
- B. The practice of understanding key species' roles and maintaining ecosystem services critical to humans and ecosystems.
- C. The process of eliminating invasive species to ensure that native species continue to dominate the ecosystem.
- D. The focus on preserving only top predators to maintain ecological balance and prevent ecosystem collapse

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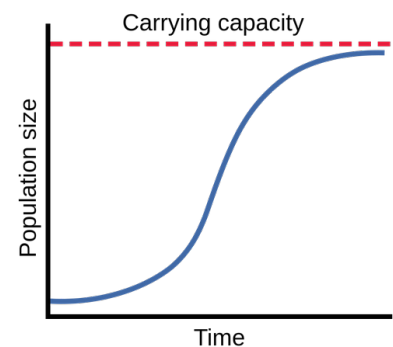
It's OK if you don't know this if you're not in Ms. Lerner's classes!

13. Which of the following are actions that could be taken to directly alleviate habitat loss?

- A. Building wildlife corridors
- B. Building riparian corridors
- C. Both A and B
- D. None of the above

14. Which of the following is TRUE in regard to the given graph?
(choose the best answer)

- A. This graph depicts population levels over time in an unrealistic manner.
- B. This graph depicts population levels over time in a realistic manner.
- C. Population growth could have subsided because of a high emigration rate and a very high birth rate.
- D. Population growth could have subsided because of a very low death rate and a high emigration rate.



15. The fact that Earth Overshoot Day is moving closer to the beginning of the year over time means that:

- A. We are borrowing resources from the past by decreasing resource availability to other organisms.
- B. We are increasing human carrying capacity.
- C. We are decreasing biodiversity in the present.
- D. None of the above

Section I ends.

SECTION II: FREE RESPONSE

ANSWER THE FOLLOWING FRQs TO THE BEST OF YOUR ABILITY. WRITE ON THIS EXAM.

1 QUESTION – 5 POINTS – 5 MINUTES

1. **Fill In The Blanks:** (each blank is worth 1 pt)

CHIPPO stands for Climate Change, Habitat Loss, Invasive Species, Pollution, Population, and overexploitation / overharvesting / overconsumption. (← all 3 are valid answers)

A beautiful view of a rainforest is an example of a cultural ecosystem service.

(If the word “cultural” is not included, deduct 0.5 pts)

(subtype identification req'd for Ms. Lerner's students!)

Most producers are able to perform photosynthesis to obtain energy from the sun and also perform cellular respiration to convert energy stored in biomolecules like glucose into usable energy in the form of ATP.

Section II ends.

- END OF EXAM -

Section II ends. © StudyBio (studybio.org)